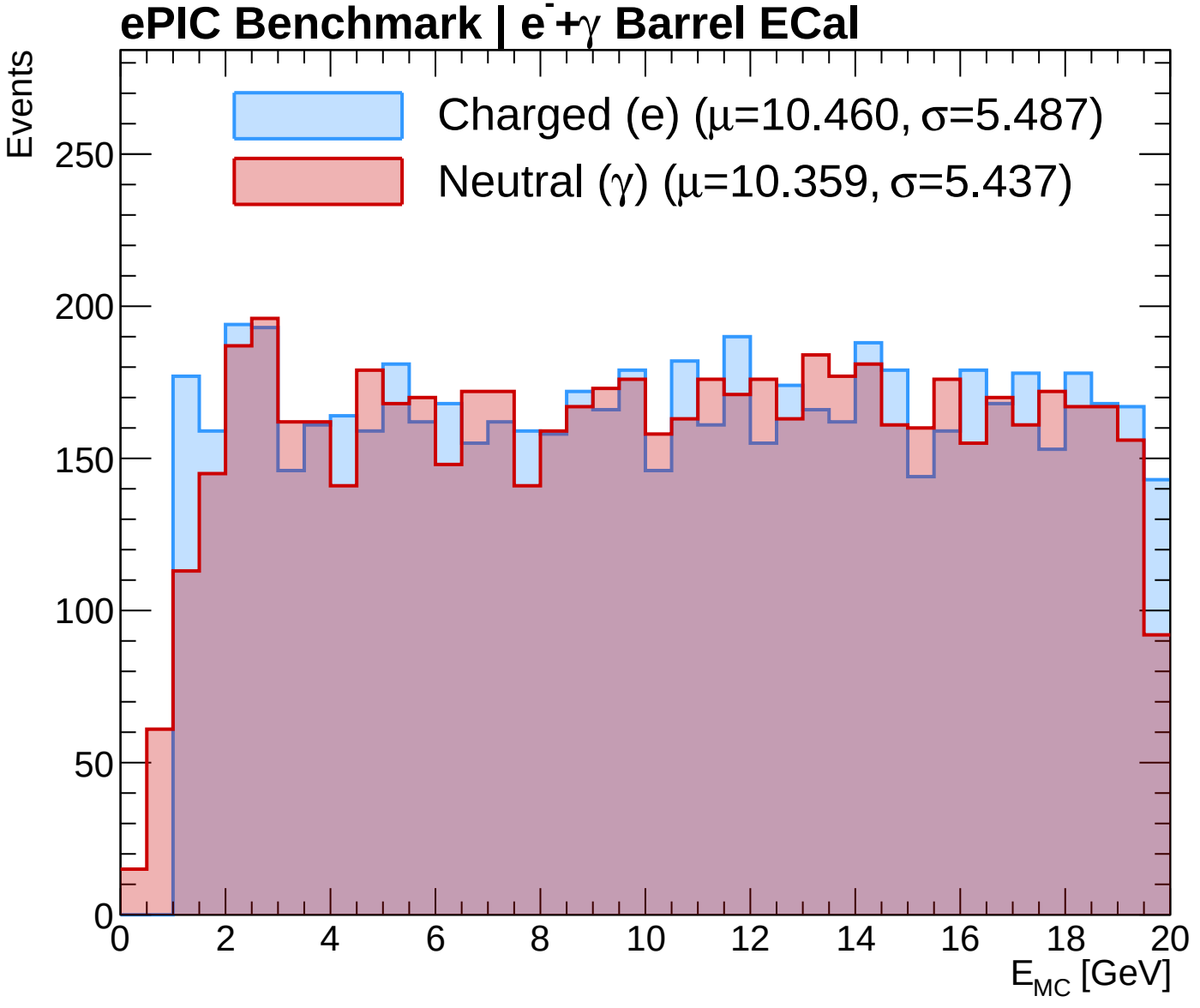
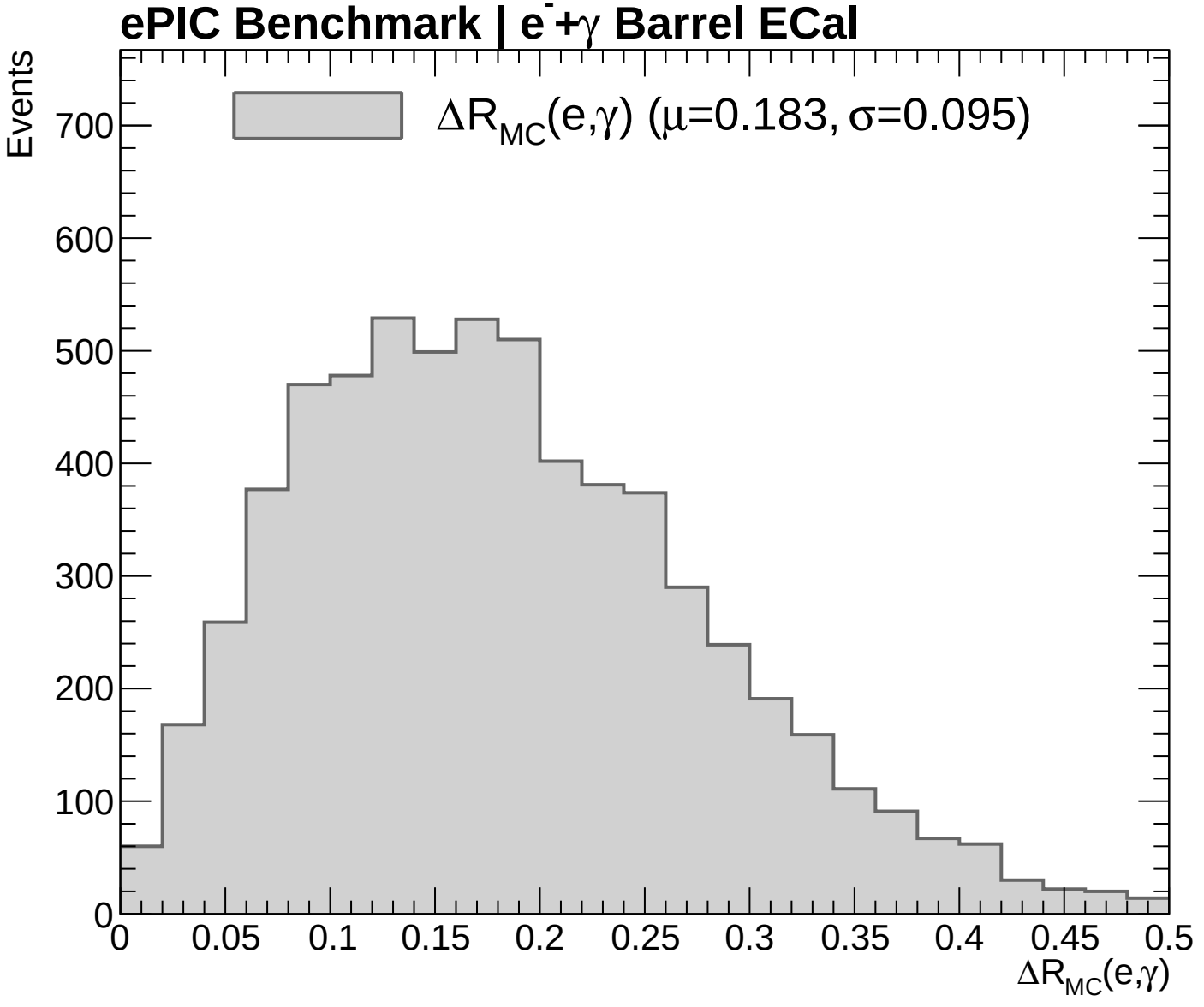
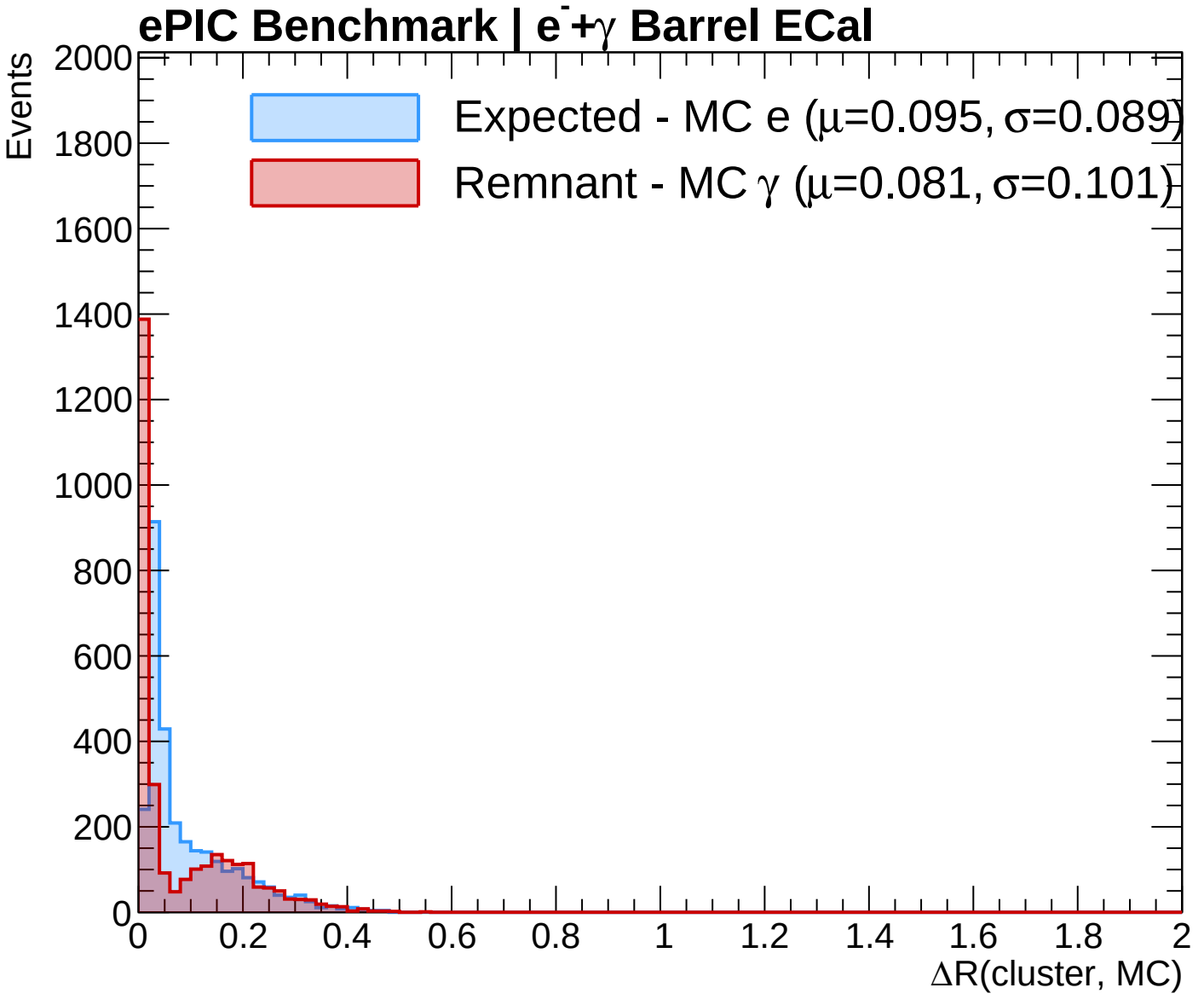
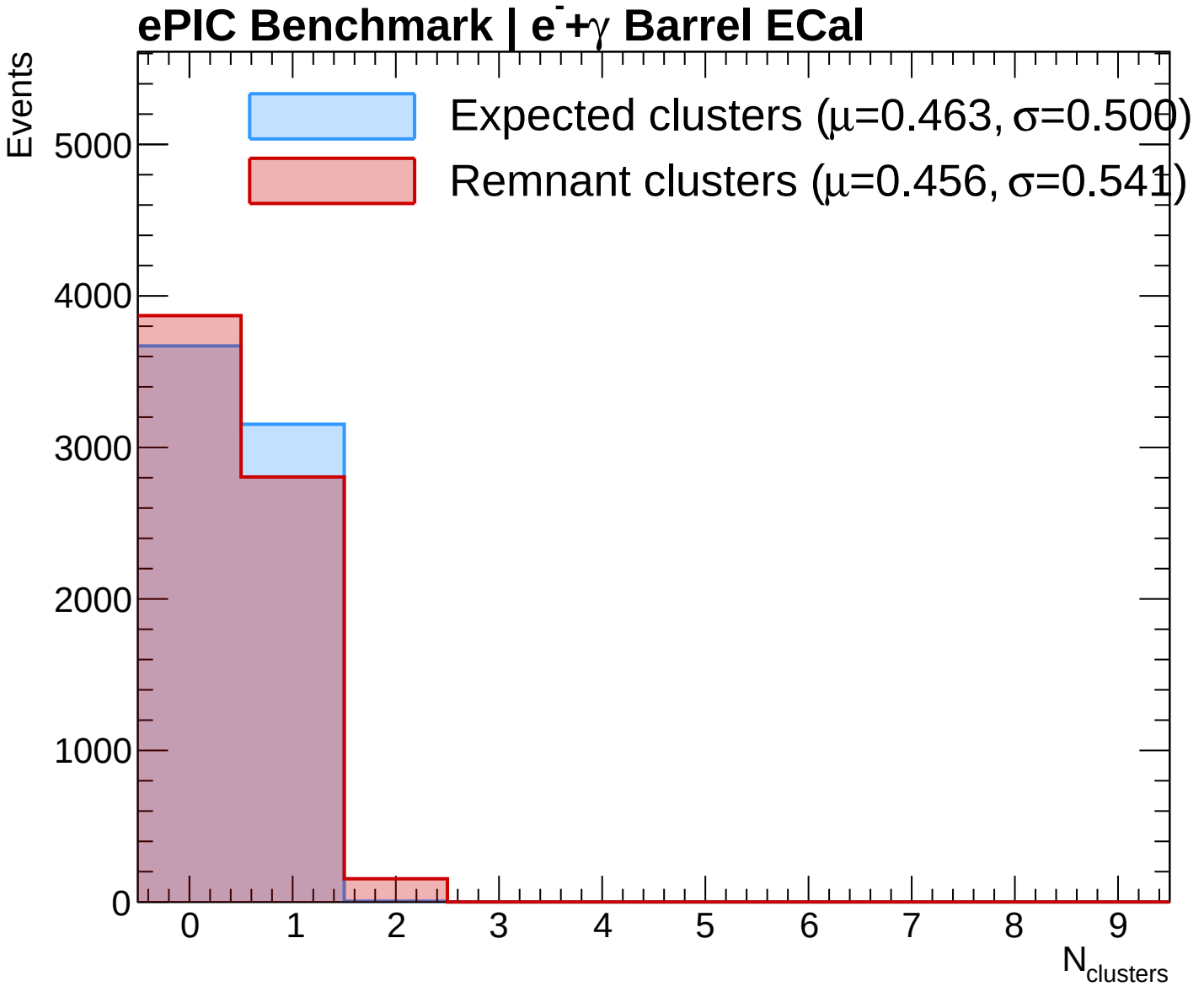
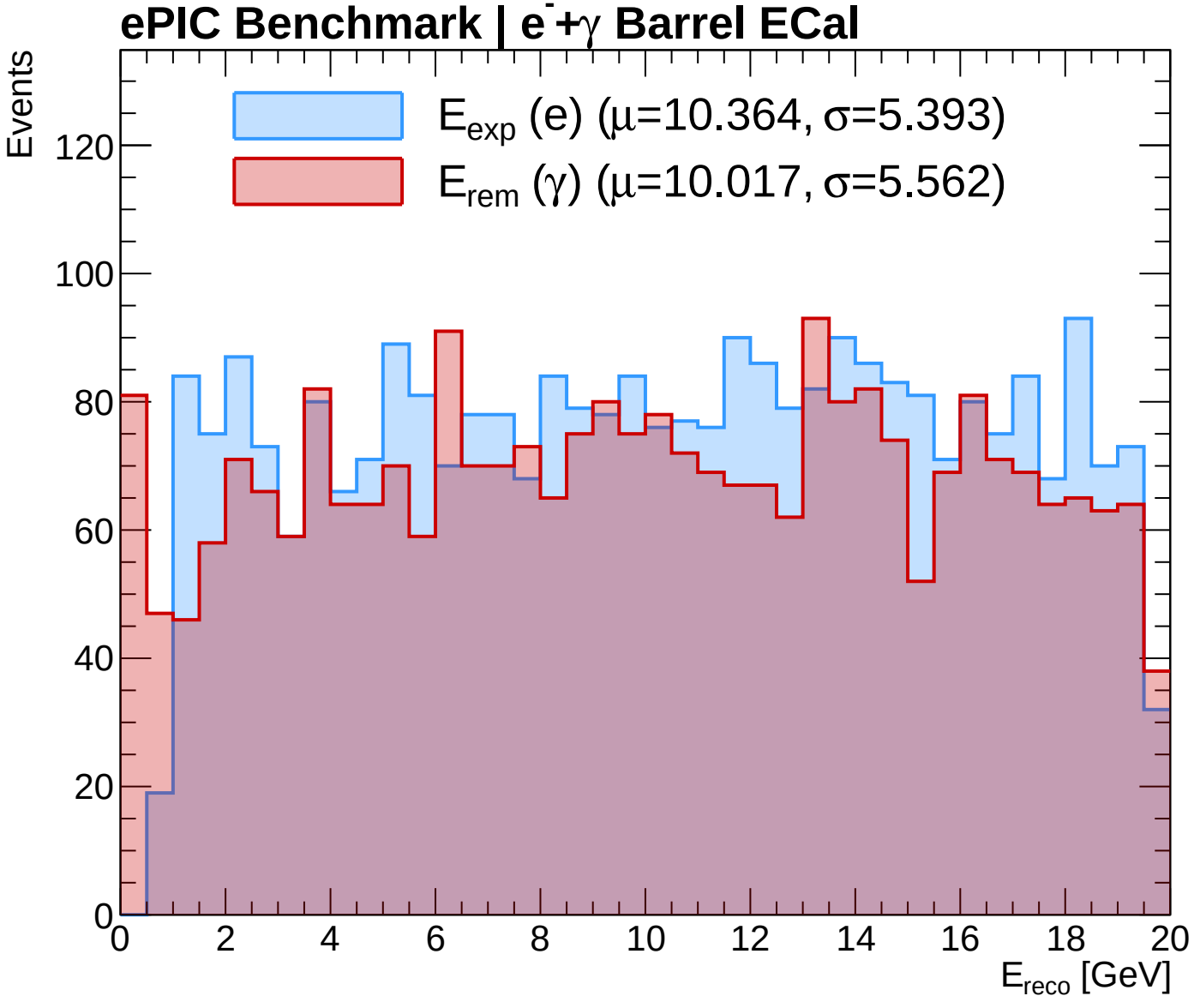
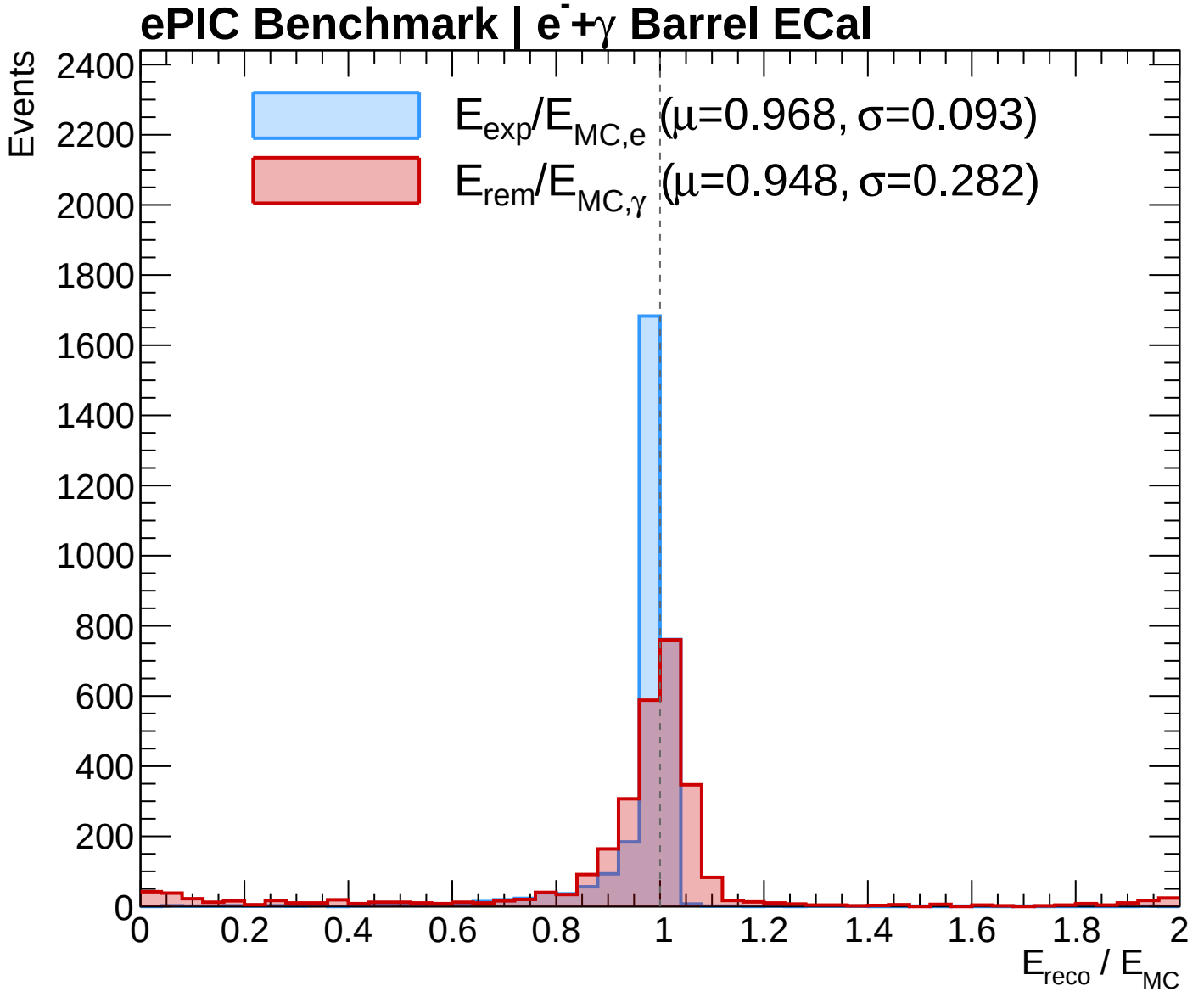

ePIC Benchmark | e⁻+γ Barrel ECal
Events
Charged (e) (μ=10.460, σ=5.487)
Neutral (γ) (μ=10.359, σ=5.437)
0
50
100
150
200
250
0
2
4
6
8
10
12
14
16
18
20
E_MC [GeV]

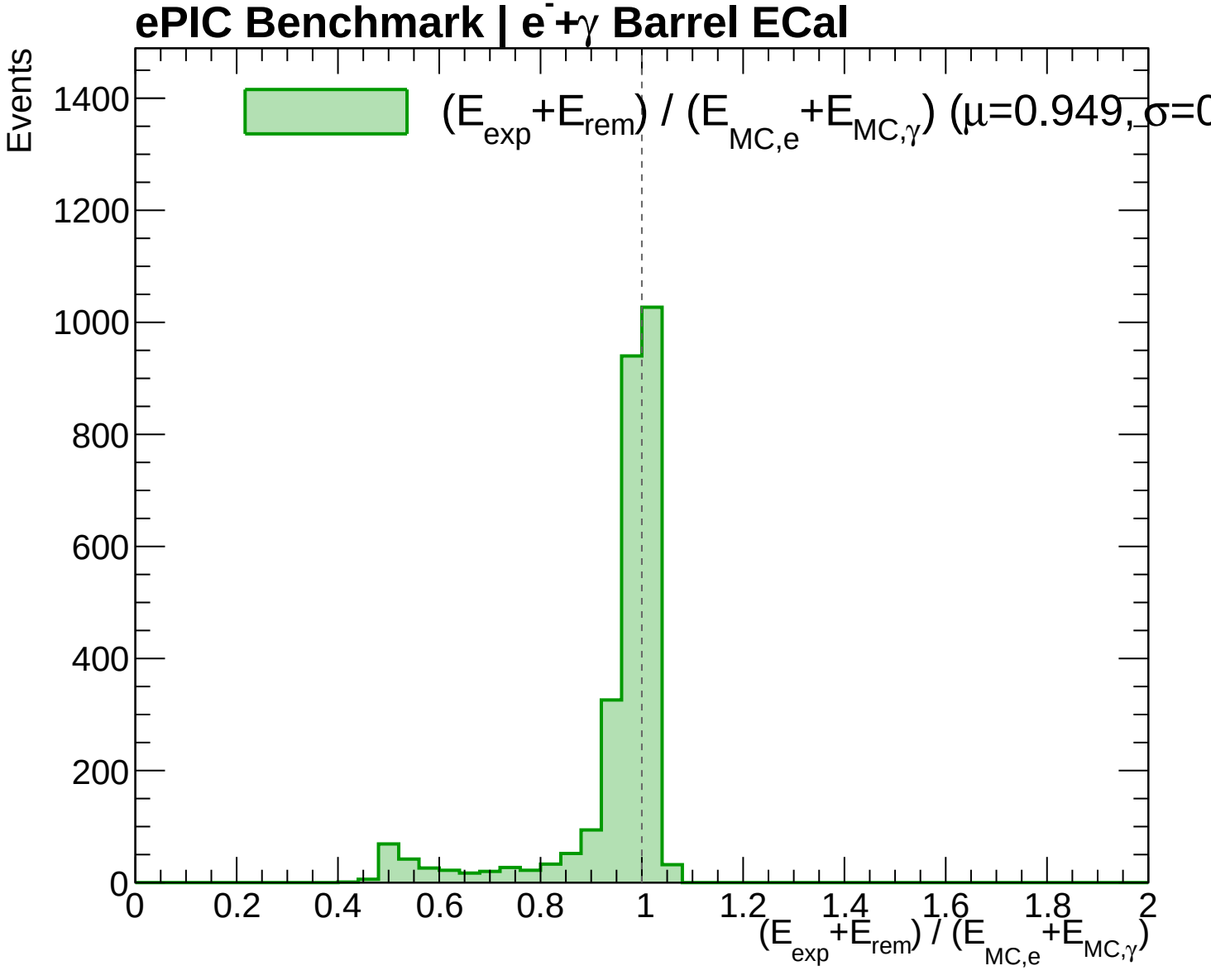

ePIC Benchmark | e⁻+γ Barrel ECal
Events
ΔR_MC(e,γ) (μ=0.183, σ=0.095)
700
600
500
400
300
200
100
0
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
ΔR_MC(e,γ)

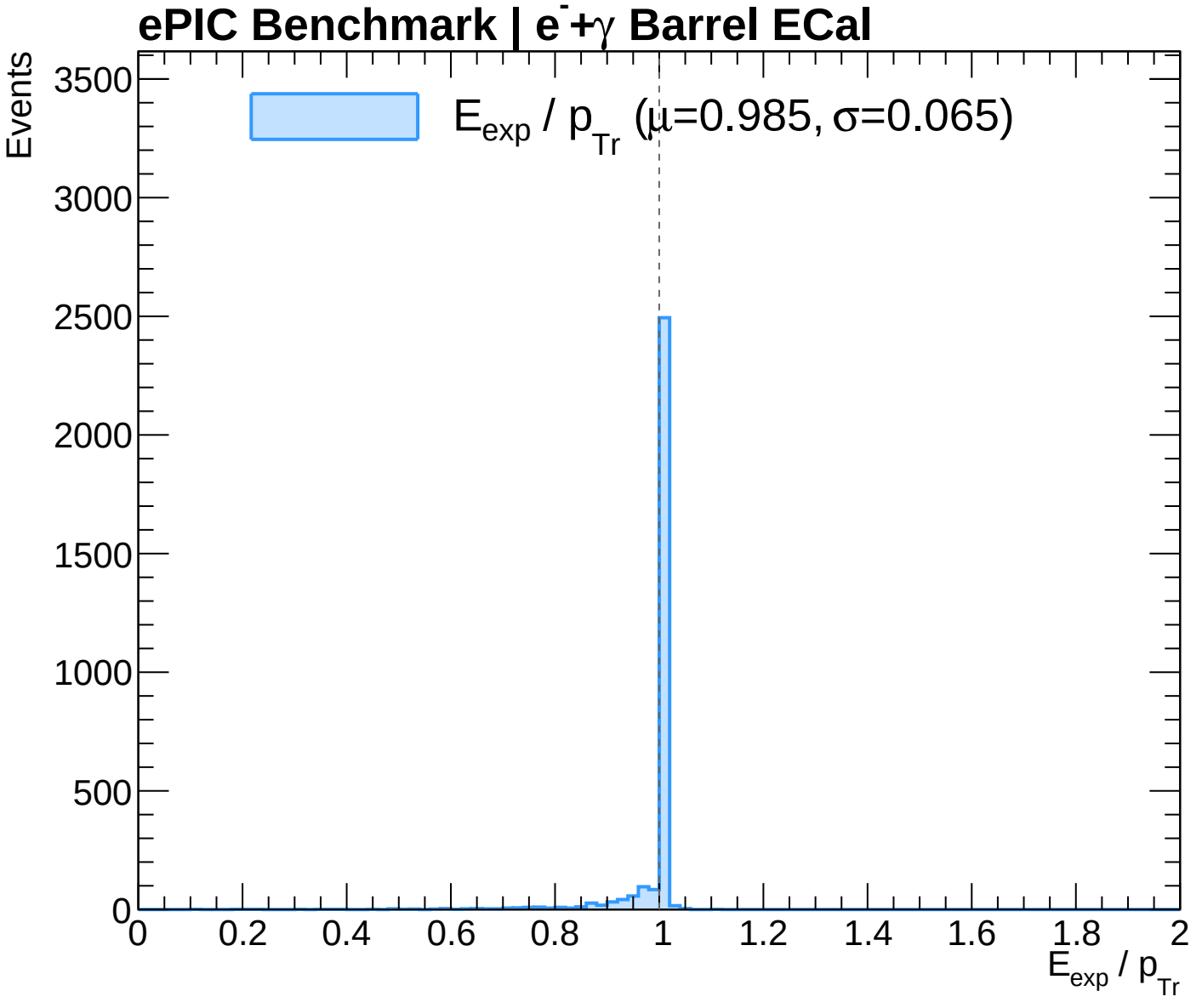

ePIC Benchmark | e⁻+γ Barrel ECal
Events
Expected - MC e (μ=0.095, σ=0.089)
Remnant - MC γ (μ=0.081, σ=0.101)
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
ΔR(cluster, MC)

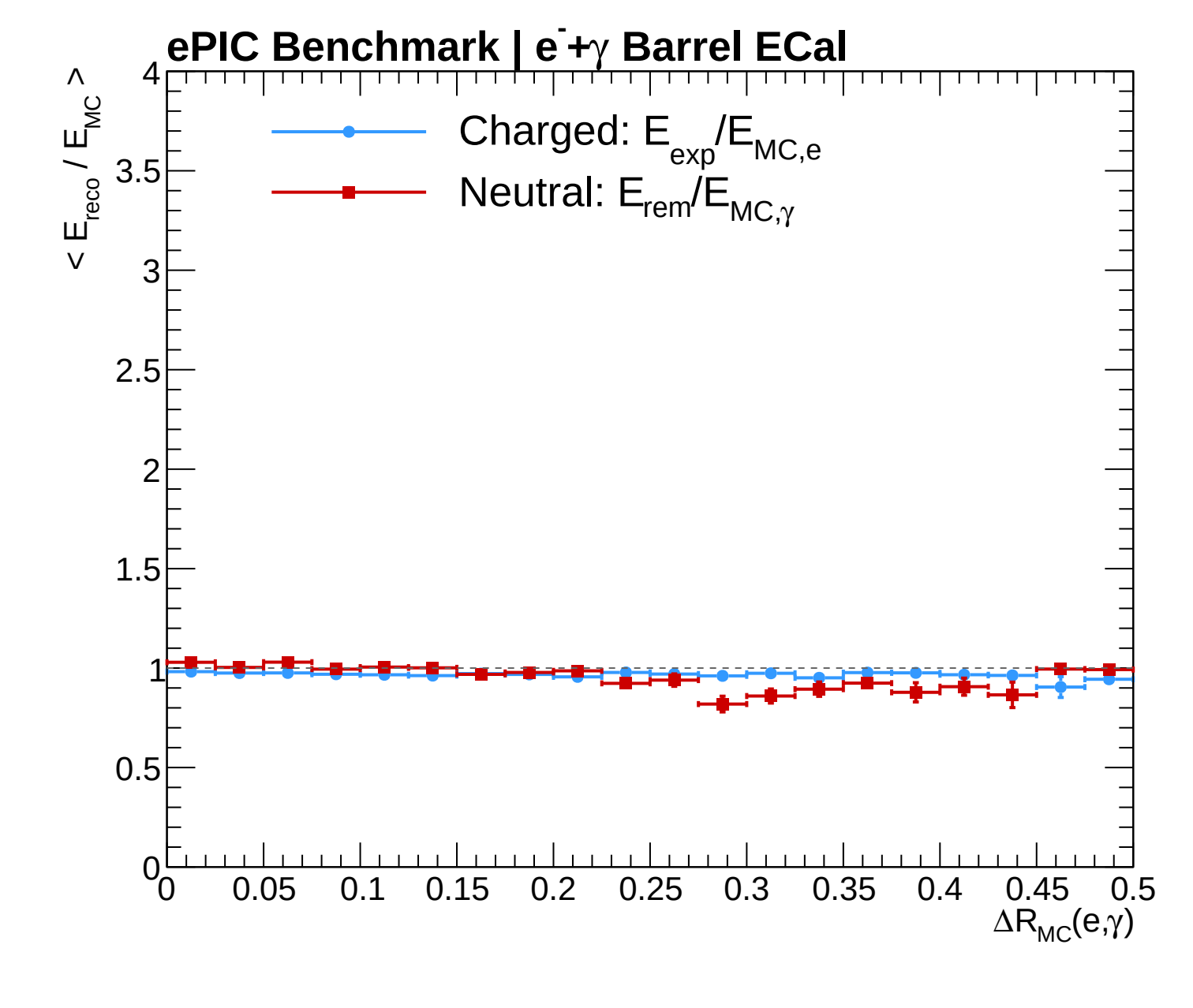

ePIC Benchmark | e⁻+γ Barrel ECal
Events
5000
4000
3000
2000
1000
0
0
1
2
3
4
5
6
7
8
9
N clusters
Expected clusters (μ=0.463, σ=0.500)
Remnant clusters (μ=0.456, σ=0.541)

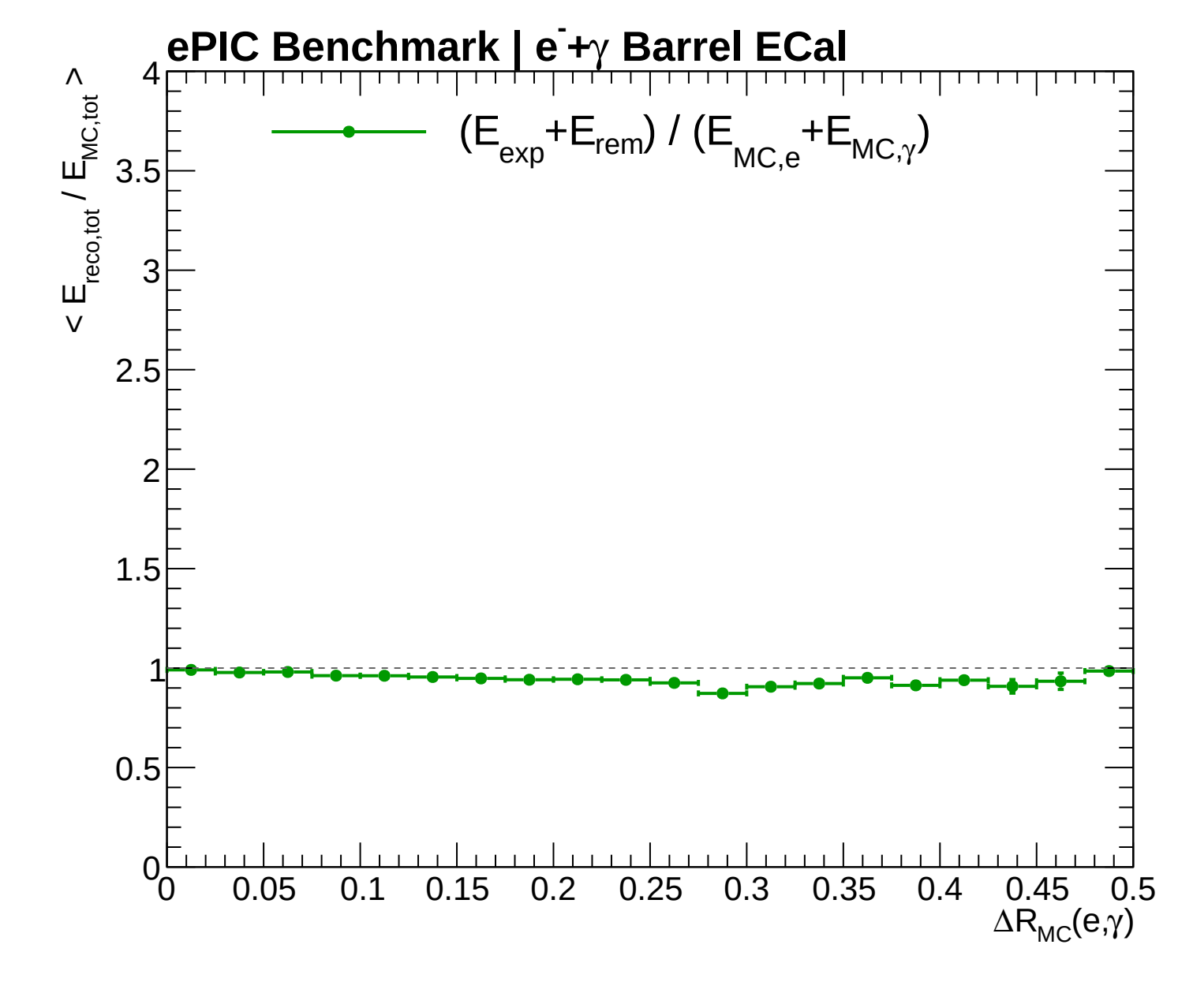

ePIC Benchmark | e⁻+γ Barrel ECal
Events
E_exp (e) (μ=10.364, σ=5.393)
E_rem (γ) (μ=10.017, σ=5.562)
0
20
40
60
80
100
120
0
2
4
6
8
10
12
14
16
18
20
E_reco [GeV]

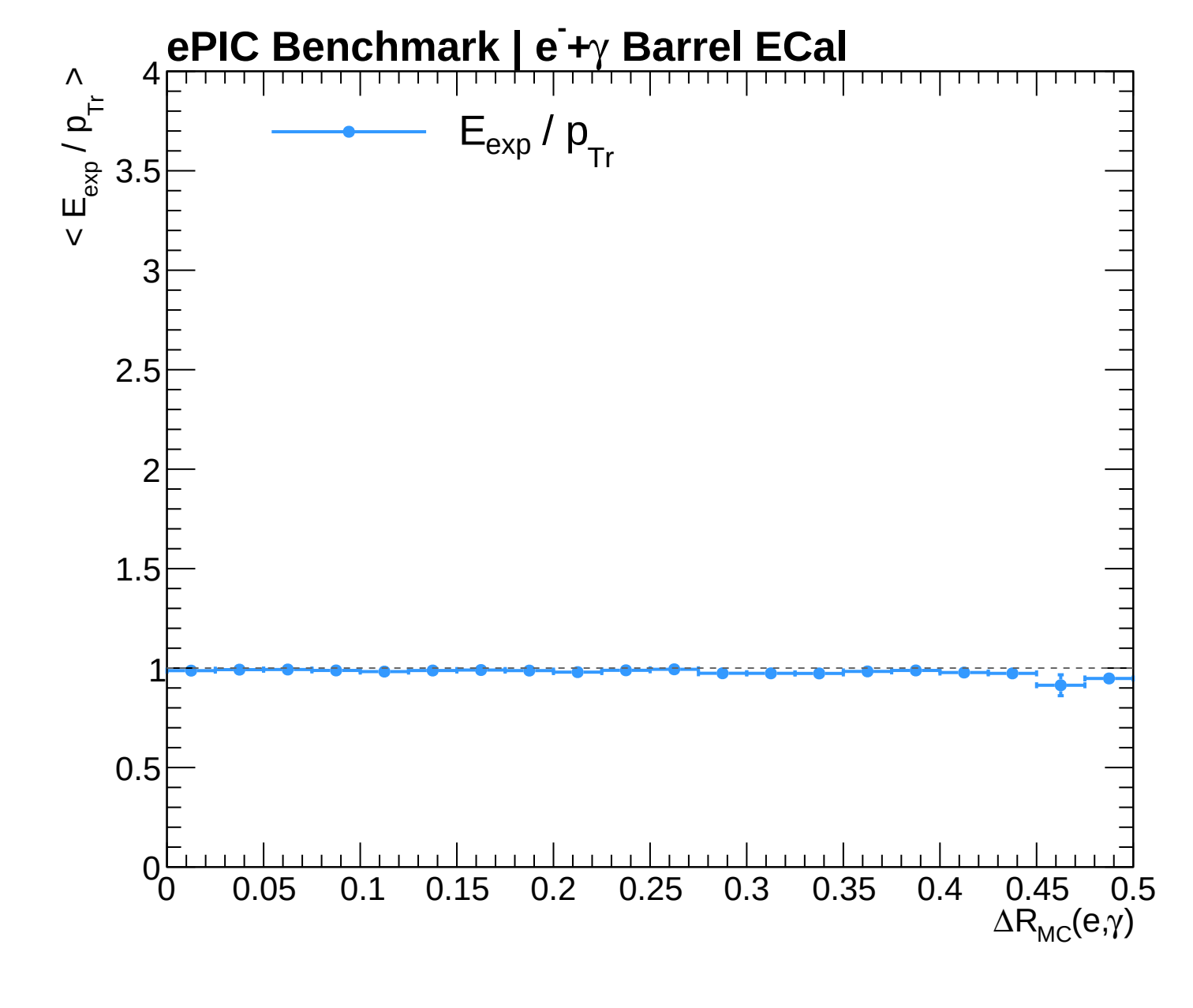

ePIC Benchmark | e⁻+γ Barrel ECal
Events
2400
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
E_exp/E_MC,e (μ=0.968, σ=0.093)
E_rem/E_MC,γ (μ=0.948, σ=0.282)
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_reco / E_MC

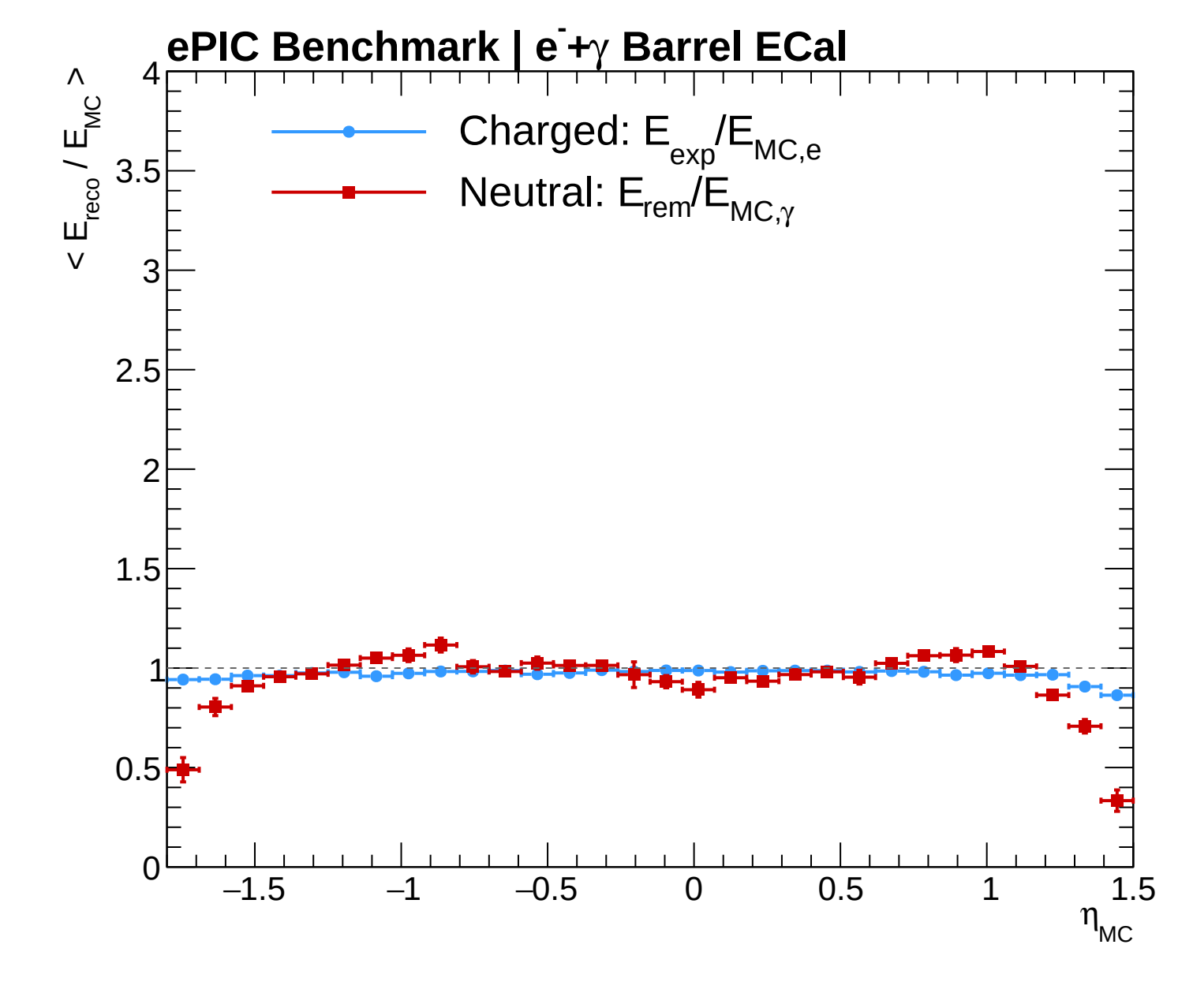

ePIC Benchmark | e⁻+γ Barrel ECal
Events
(E_exp+E_rem) / (E_MC,e+E_MC,γ) (μ=0.949, σ=0
0
200
400
600
800
1000
1200
1400
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
(E_exp+E_rem) / (E_MC,e+E_MC,γ)

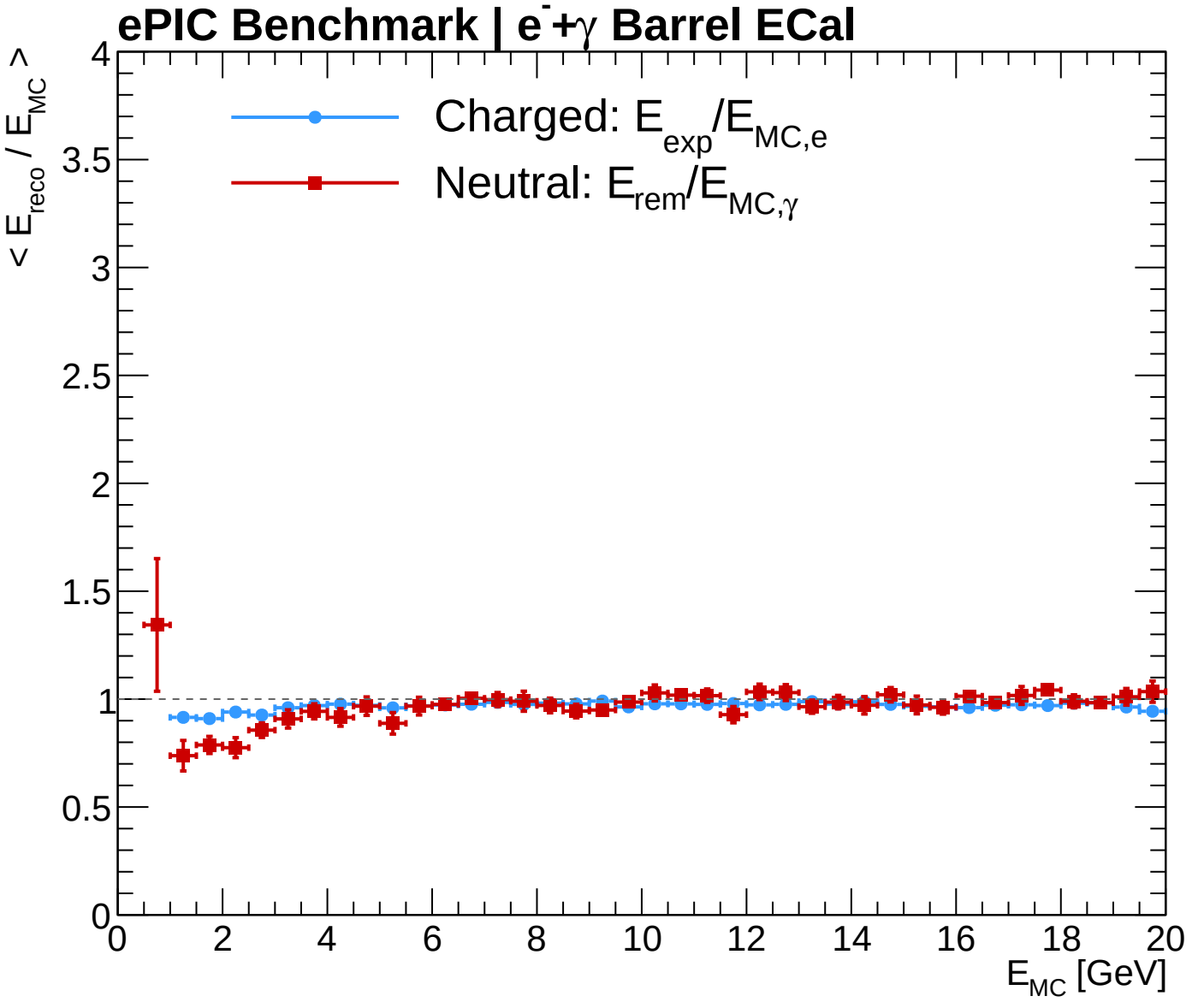

ePIC Benchmark | e⁻+γ Barrel ECal
Events
3500
3000
2500
2000
1500
1000
500
0
E_exp / p_Tr (μ=0.985, σ=0.065)
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_exp / p_Tr

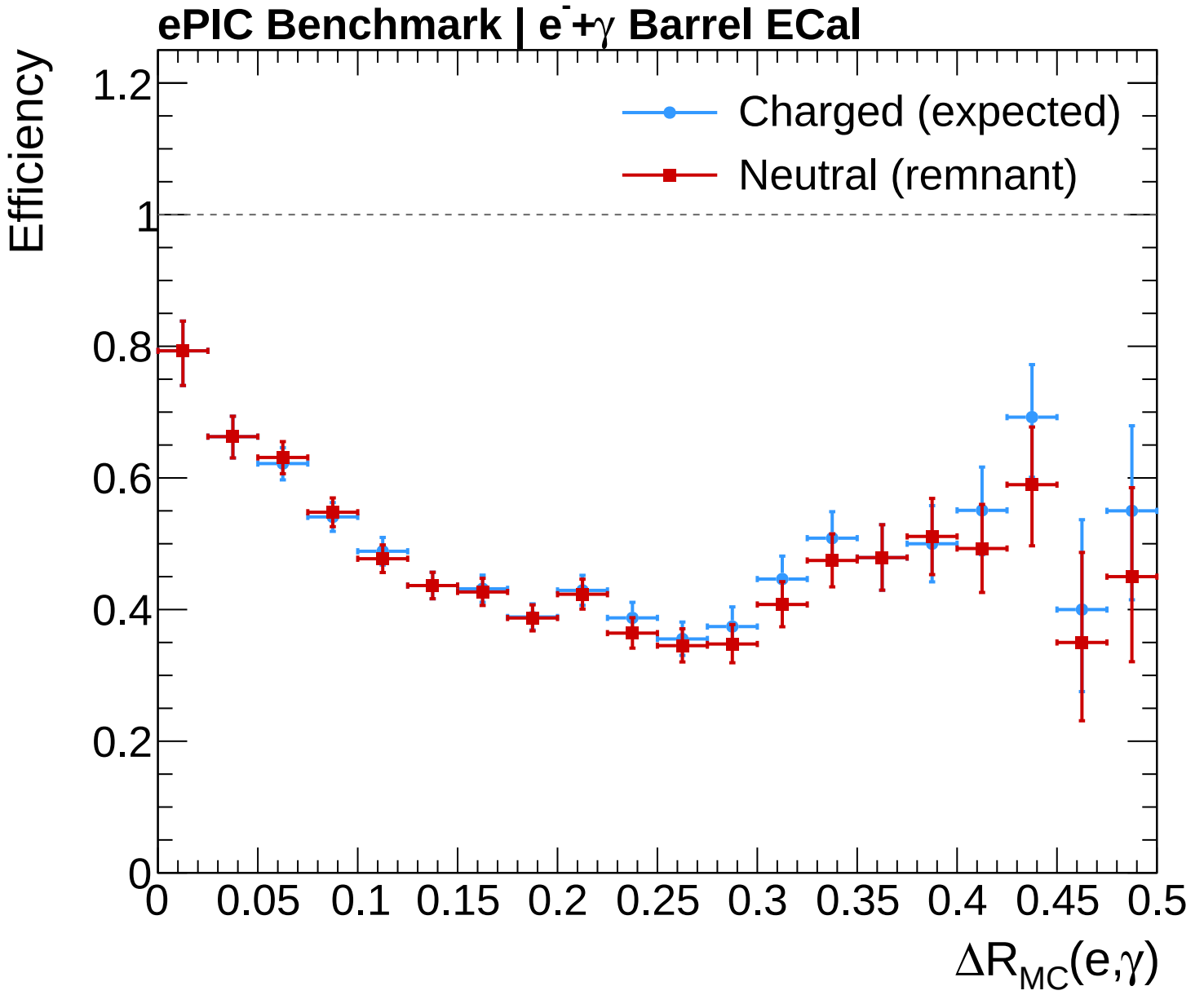

ePIC Benchmark | e⁻+γ Barrel ECal
Charged: E_exp/E_MC,e
Neutral: E_rem/E_MC,γ
< E_reco / E_MC >
ΔR_MC(e,γ)
0
0.5
1
1.5
2
2.5
3
3.5
4
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5

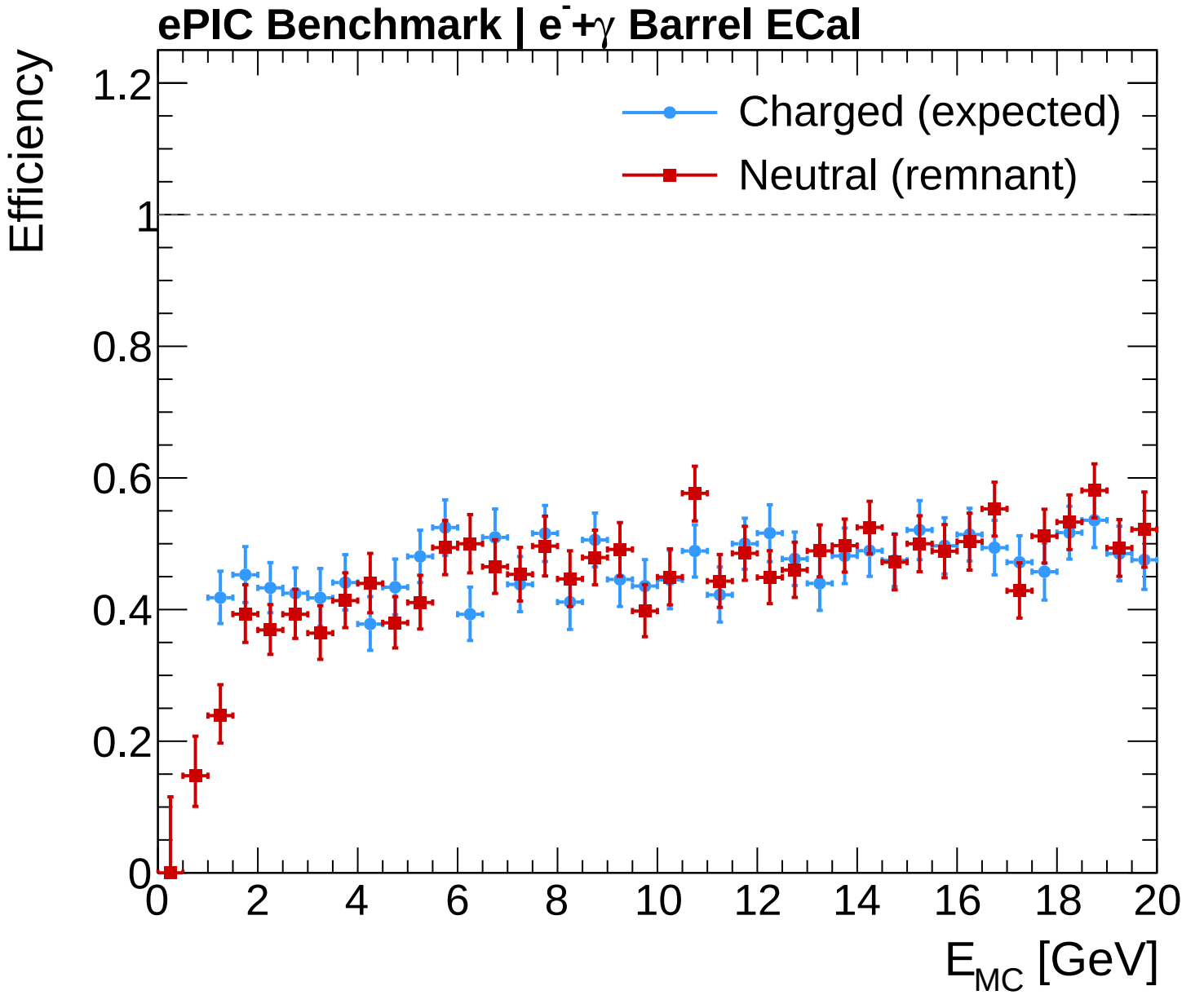

ePIC Benchmark | e⁻+γ Barrel ECal
(E_exp+E_rem) / (E_MC,e+E_MC,γ)
< E_reco,tot / E_MC,tot >
ΔR_MC(e,γ)
0
0.5
1
1.5
2
2.5
3
3.5
4
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5

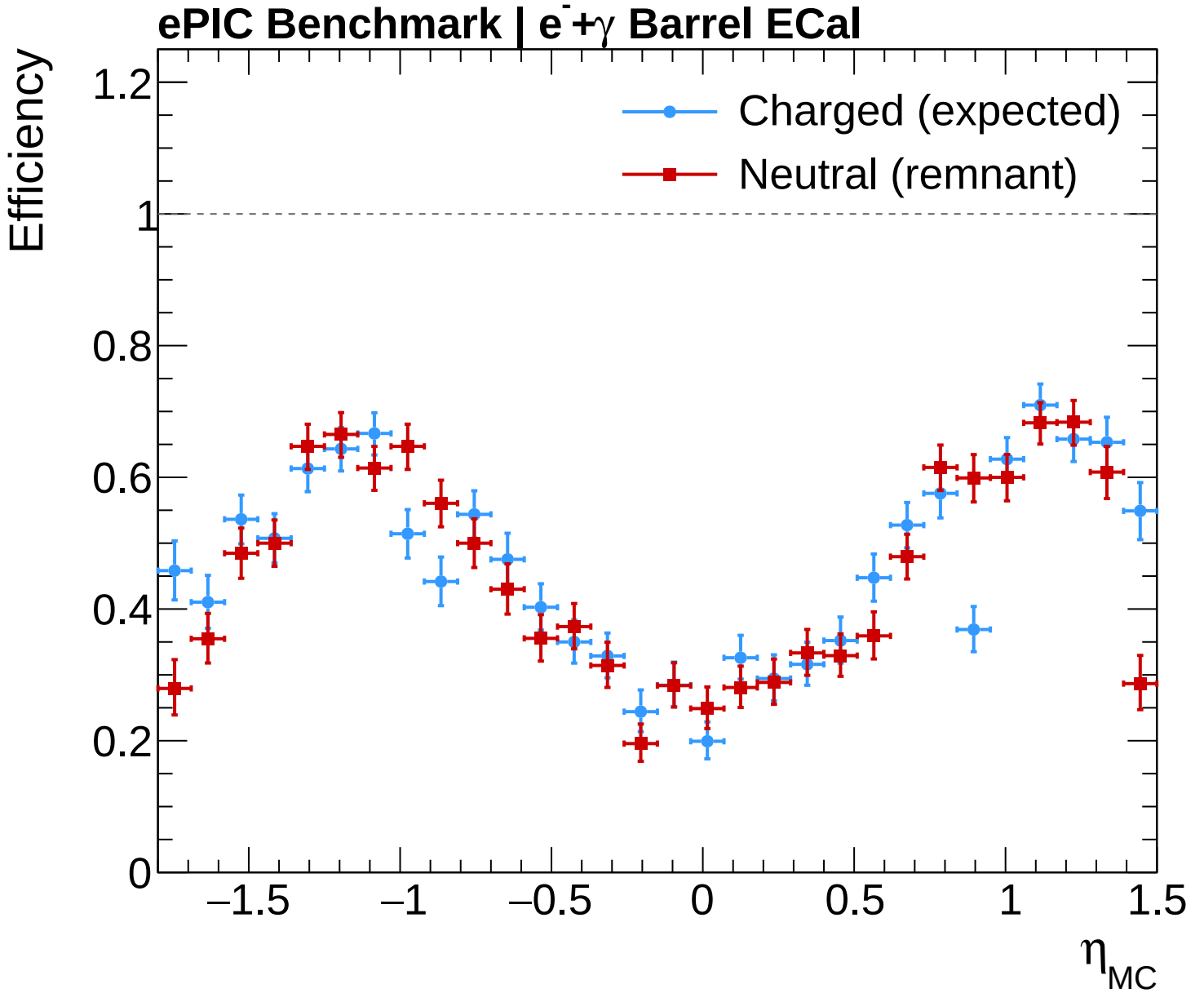

ePIC Benchmark | e⁻+γ Barrel ECal
< E_exp / p_Tr >
E_exp / p_Tr
4
3.5
3
2.5
2
1.5
1
0.5
0
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
ΔR_MC(e,γ)


ePIC Benchmark | e⁻+γ Barrel ECal
Charged: E_exp/E_MC,e
Neutral: E_rem/E_MC,γ
< E_reco / E_MC >
η_MC
4
3.5
3
2.5
2
1.5
1
0.5
0
–1.5
–1
–0.5
0
0.5
1
1.5


ePIC Benchmark | e⁻+γ Barrel ECal
Charged: E_exp/E_MC,e
Neutral: E_rem/E_MC,γ
< E_reco / E_MC >
E_MC [GeV]
0
0.5
1
1.5
2
2.5
3
3.5
4
0
2
4
6
8
10
12
14
16
18
20


ePIC Benchmark | e⁻+γ Barrel ECal
Efficiency
Charged (expected)
Neutral (remnant)
1.2
1
0.8
0.6
0.4
0.2
0
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
ΔR_MC(e,γ)


ePIC Benchmark | e⁻+γ Barrel ECal
Efficiency
Charged (expected)
Neutral (remnant)
1.2
1
0.8
0.6
0.4
0.2
0
0
2
4
6
8
10
12
14
16
18
20
E_MC [GeV]


ePIC Benchmark | e⁻+γ Barrel ECal
Efficiency
Charged (expected)
Neutral (remnant)
1.2
1
0.8
0.6
0.4
0.2
0
–1.5
–1
–0.5
0
0.5
1
1.5
η_MC